

ATMORE AGDM, AB, Canada

WMO: 712870

Lat: 54.7802N

Lon: 112.8248W

Elev: 1883

StdP: 13.72

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-28.2	-21.1	-34.6	0.8	-28.0	-27.1	1.3	-20.4	22.5	35.3	18.9	29.5	2.7	230	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.2	82.0	64.4	78.8	62.3	75.8	61.1	66.8	78.6	64.6	75.4	62.7	72.7	7.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	90.0	72.1	60.3	83.7	69.6	58.5	78.6	67.5	32.4	78.9	30.6	75.4	29.2	72.7	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	16.5	14.2	-38.2	87.8	6.1	3.4	-42.6	90.2	-46.2	92.2	-49.6	94.1	-54.1	96.5
			-38.0	70.7	6.1	2.3	-42.4	72.3	-46.0	73.6	-49.4	74.9	-53.8	76.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	10.1	11.0	23.2	37.8	49.8	57.1	61.5	58.4	49.7	37.8	21.4	9.1
	DBStd	21.82	15.63	14.49	13.09	9.31	7.85	5.29	4.90	5.85	7.67	8.37	12.17	14.35
	HDD50	6259	1236	1092	831	375	101	8	0	6	97	385	859	1269
	HDD65	10730	1701	1512	1296	816	474	242	131	214	459	842	1309	1734
	CDD50	1044	0	0	0	9	93	220	358	267	89	8	0	0
	CDD65	40	0	0	0	0	1	5	23	10	1	0	0	0
	CDH74	1042	0	0	0	2	79	162	455	259	78	7	0	0
	CDH80	185	0	0	0	0	12	19	101	40	13	0	0	0
	WSAvg	6.1	5.7	5.7	6.5	7.4	7.2	6.4	5.9	5.7	6.1	6.3	5.7	5.3
	PrecAvg	15.9	0.7	0.5	0.6	0.9	1.6	2.8	3.1	2.0	1.3	0.7	0.6	0.7
(15) Precipitation	PrecMax	20.0	1.5	1.1	1.3	2.2	3.4	5.4	6.1	3.5	2.9	2.2	1.6	1.5
	PrecMin	11.8	0.0	0.0	0.1	0.1	0.2	0.9	0.9	0.8	0.4	0.2	0.2	0.0
	PrecStd	2.4	0.4	0.3	0.4	0.5	0.7	1.2	1.1	0.7	0.7	0.4	0.4	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.0	45.2	53.4	72.2	81.4	82.6	87.5	84.4	82.0	72.7	51.7	43.4
		MCWB	37.3	37.1	42.5	51.4	57.7	63.2	67.8	65.2	62.4	54.1	43.9	36.7
	2%	DB	39.8	40.5	47.9	64.7	75.4	78.6	82.7	80.2	75.4	61.8	44.4	38.1
		MCWB	35.1	34.6	39.0	47.8	54.9	61.0	66.2	64.6	59.6	49.6	38.1	33.5
	5%	DB	36.1	36.0	44.1	59.3	71.1	75.0	78.8	76.6	70.4	56.2	40.1	33.1
		MCWB	32.7	31.5	36.7	44.8	53.0	59.5	63.9	62.6	56.8	45.5	35.7	30.0
	10%	DB	32.1	32.1	40.4	54.5	67.4	71.3	75.6	73.3	64.9	51.6	36.0	28.6
		MCWB	29.5	29.2	34.2	42.2	50.9	57.8	62.6	61.2	54.2	43.2	32.7	26.6
	0.4%	WB	37.9	37.8	43.0	52.2	59.2	65.8	70.6	69.1	63.6	55.0	43.9	37.2
		MCDB	42.8	43.7	52.1	70.6	75.8	79.0	82.2	79.7	78.4	70.6	51.5	42.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.6	34.9	39.6	48.3	56.6	63.1	67.7	66.0	60.4	50.2	39.3	33.5
		MCDB	39.5	40.0	47.0	63.3	72.2	74.1	80.4	77.3	72.8	59.6	43.0	37.4
	5%	WB	32.8	32.2	37.1	45.4	54.4	61.2	65.6	63.7	57.7	46.6	35.9	30.1
		MCDB	36.1	35.7	43.5	57.7	68.2	71.8	76.4	74.2	68.3	54.6	39.5	32.8
	10%	WB	29.3	29.0	34.6	42.8	52.2	59.1	63.7	61.9	55.1	44.0	32.8	26.7
		MCDB	32.0	32.4	39.8	53.3	64.4	68.9	73.3	71.7	63.3	50.5	35.7	28.6
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	22.1	21.0	21.9	25.3	23.3	24.2	24.5	23.7	19.6	16.1	17.5
		MCDBR	19.3	25.5	23.0	31.1	32.6	30.3	30.2	30.3	33.2	28.9	20.1	20.4
		MCWBR	15.2	20.3	15.3	17.1	15.8	15.0	15.5	15.8	18.0	17.2	14.3	16.7
	5% WB	MCDBR	18.9	24.6	22.0	29.2	29.4	26.2	27.5	27.4	30.9	26.8	18.7	19.8
		MCWBR	15.1	20.0	14.7	16.7	14.7	14.1	15.0	15.1	17.3	16.5	13.8	16.5
(40) Clear-Sky Solar Irradiance	taub		0.250	0.275	0.301	0.330	0.342	0.355	0.357	0.365	0.313	0.283	0.252	0.234
	taud		2.307	2.307	2.348	2.390	2.413	2.381	2.435	2.355	2.518	2.571	2.465	2.301
	Ebn at Noon		227	260	279	285	286	282	280	270	274	260	231	209
	Edh at Noon		17	24	30	33	34	36	33	34	25	19	15	13
(44) All-Sky Solar Radiation	RadAvg		254	571	1045	1465	1787	1856	1868	1535	1059	586	269	182
	RadStd		28	42	84	116	159	127	100	110	108	83	30	14

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page